



Pediatric Asthma Exacerbation: Outpatient Setting

VIRTUAL CARE

- q Recommend video visit if possible
- q If no improvement with 6 puffs in 2 hours at home or in 24 hours with q4h puffs/nebulizer treatment, the patient needs an in-person evaluation
- q If lethargic, unable to speak in sentences, or rapidly worsening, **call 911**

Inclusion:

> Age 1 year

Dyspnea (tachypnea in young children) in patient with known or suspected asthma

Exclusion:

Dx of viral bronchiolitis, pneumonia, or croup, history of CF, chronic lung disease, congenital cardiac disease, or airway abnormalities

Monitor: Vital signs, O2 sat

Identify High-Risk patient such as prior intubation, hospitalizations or ED visits

Medication		Dosage	
Albuterol	< 6 yrs	2.5 mg INH	2 - 6 puffs MDI
	≥ 6 yrs	2.5 – 5 mg INH	4 - 10 puffs MDI
Ipratropium (Atrovent)	< 6 yrs	0.25 mg INH	
	≥ 6 yrs	0.25 – 0.5 mg INH	
Dexamethasone IM, IV, or PO		0.6mg/kg (max 16mg) – IV form OK to give PO	Discharge with 2 nd dose of 0.6mg/kg
Prednisolone	<2 yrs	2mg/kg (max 20mg)	Discharge with total 5 day course 2mg/kg/day ÷ BID
	2 – 5 yrs	2mg/kg (max 30mg)	
	6 – 11 yrs	2mg/kg (max 40mg)	
	12 + yrs	2mg/kg (max 50mg)	
Epinephrine SQ autoinj	< 25 kg	0.15 mg	Only for anaphylaxis
	≥ 25 kg	0.3 mg	

MILD Exacerbation:

- Normal mental status
- Expiratory wheezing only with minimal work of breathing, tachypnea, retractions, and speaking in full sentences

Administer:

- Albuterol INH or neb
- If using inhaler, use with aerochamber and appropriate size mask
- Can repeat Q20 min x 3
- If no improvement after second treatment or if high-risk patient, give steroids and follow moderate pathway

MODERATE to SEVERE Exacerbation:

- Dyspnea at rest
- Interfering with usual activities
- Wheeze heard throughout OR decrease aeration
- Increase work of breathing moderate

Target O2 sat >90-92%

Administer:

- Steroids immediately
- Albuterol and Atrovent INH or neb Q20 min x 3
- If using inhaler, use with aerochamber and appropriate size mask

CRITICAL Exacerbation:

- Lethargy
- Obtunded
- Unable to repeat a short phrase
- Grunting respirations
- Severe retractions
- Bradypnea, apnea
- Absent breath sounds

Call 911

Administer:

- Steroids immediately
- Albuterol/Ipratropium x Q20 min x 3 → followed by
- Albuterol Q1H while waiting for EMS

- Epinephrine* – IM epinephrine if acute asthma associated with anaphylaxis and angioedema

*It is not routinely indicated for asthma exacerbations without anaphylaxis

Assessment After 1st Hour

Mild Exacerbation:

Assess for discharge
Review and update asthma action plan, provide education/teaching

Start controller medication if indicated

Consider short course of oral steroid (optional if complete response to single dose of bronchodilator and no risk factors for severe exacerbation)

Arrange for follow up with PCP within one week (TAV or office visit)

Moderate to Severe Exacerbation:

Consider observation for 1 additional hour if improving
If no improvement consider transfer to ED or direct admission

Albuterol MDI/Neb every 1 hour while awaiting transfer

Discharge criteria:

- Improved with minimal to no wheezing
- Able to maintain oxygen saturation > 90-92% on room air
- No significant increased work of breathing, no respiratory distress
- Resources at home adequate, can ensure good follow up
- Arrange for follow up with PCP within one week (TAV or office visit)

Pediatric Transfer
and Advice Center:
562-658-3980 OR
8-137-3980



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